

Retail

Generative AI helps in personalising offerings and brand management, thus optimising marketing and sales activities. It enables retailers to tailor their offerings more precisely to customer demand, and supports dynamic pricing and planning.

Personalised offerings:

Generative AI enables retailers to deliver customised experiences, offerings, pricing, and planning. It also modernises the online and physical buying experience.

Dynamic pricing and planning:

It helps predict demand for different products, providing greater confidence for pricing and stocking decisions.

Other usages of generative AI in the retail industry include:

- Campaign management
- Content management
- Augmented customer support
- Search engine optimisation

Banking

Generative AI applications deliver personalised banking experiences to customers. They improve the financial simulations, developing risk analytics.

Risk mitigation and portfolio optimisation: Generative AI helps banks to build data foundations for developing risk models, identify events that are impacting the bank, mitigate that risk, and optimise portfolios.

Customer pattern analysis: It can analyse patterns in historic banking data at scale, helping relationship managers and customer representatives to identify customer preferences, anticipate needs, and create personalised banking experiences.

Customer financial planning: Generative AI can be used to automate customer service, identify trends in customer behaviour, and predict customer needs and preferences. This helps to understand the customer better and provide personalised advice.

Other usages of generative AI in the banking industry include:

- Anti-money laundering regulations
- Compliance
- Financial simulations
- Applicant simulations
- Next best action
- Risk analytics
- Fraud prevention

Insurance

The capability of analysing and processing large amounts of data by generative AI helps in accurate risk assessments and effective claims processes.

Customer support: Generative AI can provide multilingual customer service by translating customer queries and responding to them in the preferred language.

Policy management: It analyses large amounts of unstructured data related to customer policies, various policy documents, customer feedback, and social media literature for better policy management.

Claims management: Generative AI helps in analysing various claims artefacts to enhance the overall efficiency and effectiveness of claims management.

Other uses of generative AI in the insurance industry are:

- Customer communications
- Coverage explanations
- Cross-sell and up-sell products
- Accelerate the product development life cycle
- Innovation of products

Education

Generative AI offers real-time collaboration between teachers, administrators, and technology innovators.

Student enablement: Generative AI helps students who speak different languages with real-time lesson translations, and helps the visually impaired with classroom accessibility.

Student success: It offers deep analytic insights into student success, and helps teachers to make informed

decisions on how to improve results.

Telecommunications

Generative AI adoption improves operational efficiency and network performance in the telecom industry. It can be used to:

- Analyse customer purchasing patterns
- Personalise recommendations of services
- Enhance sales
- Manage customer loyalty
- Give insights into customer preferences
- Provide better data and network security, enhancing fraud detection

Public sector

The goal of most governments is to establish a digital government and provide better citizen services. Generative AI enables smart cities and optimised service operations.

Smart cities: Generative AI helps in toll management, traffic optimisation, and sustainability.

Better citizen services: It offers easier access to connected government services through tracking, search, and conversational bots.

Other services that are enabled using generative AI are:

- Service operations optimisation
- Contact centre automation

Limitations of current LLMs

Enterprises face several challenges in implementing LLMs as a part of generative AI solutions.

Data preparation: Identifying data sources for LLM, labelling data for algorithms, data policies, data security, data storage and data governance are the big challenges.

High volume data: LLMs reference huge amounts of data to generate meaningful output. This training and deploying of LLMs require significant computational resources, making them inaccessible to smaller enterprises or researchers with limited budgets.